Northeast Fisheries Science Center
Makara Data Submission Guide

Version 1.2

National Oceanic and Atmospheric Administration
National Marine Fisheries Service
Northeast Fisheries Science Center
Passive Acoustics Branch
Woods Hole, Massachusetts
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General Information

This guide provides data contributors with instructions, field definitions, and examples for how to enter acoustic data and metadata into the template tables and submit them for archiving in the Northeast Fisheries Science Center's (NEFSC) Makara Database. Data that are archived in Makara may also be displayed on the Passive Acoustic Cetacean Map (PACM), with permission from the data owner(s).

Document History and Revisions

Version	Date	Author	Description
1.0	May 1, 2025	Julianne Wilder	Document Created
1.1	June 10, 2025	Jeffrey Walker	Updated "Data Table Template Instructions" to clarify that filenames and folder structure must match "templates" sub-directory.
1.2	August 26, 2025	Jeffrey Walker	• Renamed the **organization_code** to **deployment_organization_code** in the analyses and detections tables to clarify that this field refers to the organization that owns the deployment. Also added **analysis_organization_code** to the detections table. Both tables now include both the **deployment_organization_code** (organization that the deployment and recording(s) belong to), and **analysis_organization_code** (organization that the analysis belongs to). • Added **track_uri** to tracks table for linking tracks to external files in cloud storage (useful for linking track metadata to high resolution data not stored in makara) • Fixed minor typos

Contact Information for the Makara Database Support Team

If you have questions or issues regarding your data submission, please contact nmfs.nec.pamdata@noaa.gov.

Tips and Recommendations

- Timezones/Timestamps: All timestamps that are entered into the data submission templates should include the timezone (even if they are in UTC; e.g., 2023-10-12T18:09:05+0000). The NEFSC generally recommends that all recordings are made in UTC. Other timezones are okay, so long as the timezone information is included within the timestamp (e.g., 2023-10-12T18:09:05+0500 to indicate that the audio was recorded in Eastern Standard Time).

- Scenarios and Examples: It is highly recommended that data contributors look through our Scenarios section and the 'examples' template folder in the downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage prior to entering their data into the data submission templates. There may be an example that is similar to their situation.

Data Table Template Instructions

To submit data and metadata to the Makara database, the information must be organized into template tables. The data template tables and example tables are provided in a downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage, and are described in more detail below. Some of the tables are required, but do not necessarily need to be submitted at the same time. Other tables are optional, but it is highly recommended that the data contributor fills out any tables that are applicable to their deployment(s). Within each table, there are required and optional (but recommended if applicable) fields. The required tables/fields are the *bare minimum* of what metadata should be submitted.

Within the template zipped folder is a sub-directory named “templates/”. This sub-directory contains empty CSV files that can be used as starting templates for filling out each table. Each data submission should contain one or more of these CSV files with data rows properly entered based on the formats described below. The names of each CSV file as well as the folder structure must match what is in the templates folder. Data cannot be submitted as an Excel Workbook (*.xlsx file).

Required Data Tables

The following tables are required for any data submission, but do not necessarily need to be submitted at the same time (see the "When to Submit" column below). For each table, the data needs to be submitted only once unless updates or corrections are needed. In those cases, a revised table may be submitted or data contributors can edit individual fields directly in Makara.

Data Table	Definition	When to Submit
Deployments	The use of a platform (e.g., bottom-mounted mooring, buoy, glider, towed array) to record audio over a period of time	After a deployment has ended (the start date/time of a deployment is a required field, so a partial table could be submitted once a recorder is in the water, but would need to be updated after retrieval)

Recordings	Single or multichannel recordings that were made over the course of one or more deployments	After a deployment has ended
Analyses	The metadata needed to understand the detection results (i.e., the parameters of the analysis protocol)	After the data have been analyzed (if applicable, at the same time as the Detections table)
Detections	A time-binned or event-based identification of a sound source (e.g., dolphin, Atlantic cod, humpback whale, rain, vessel) produced from a passive acoustic detector or manual analysis that is used to determine presence of the sound source	After a recording has been analyzed (if applicable)

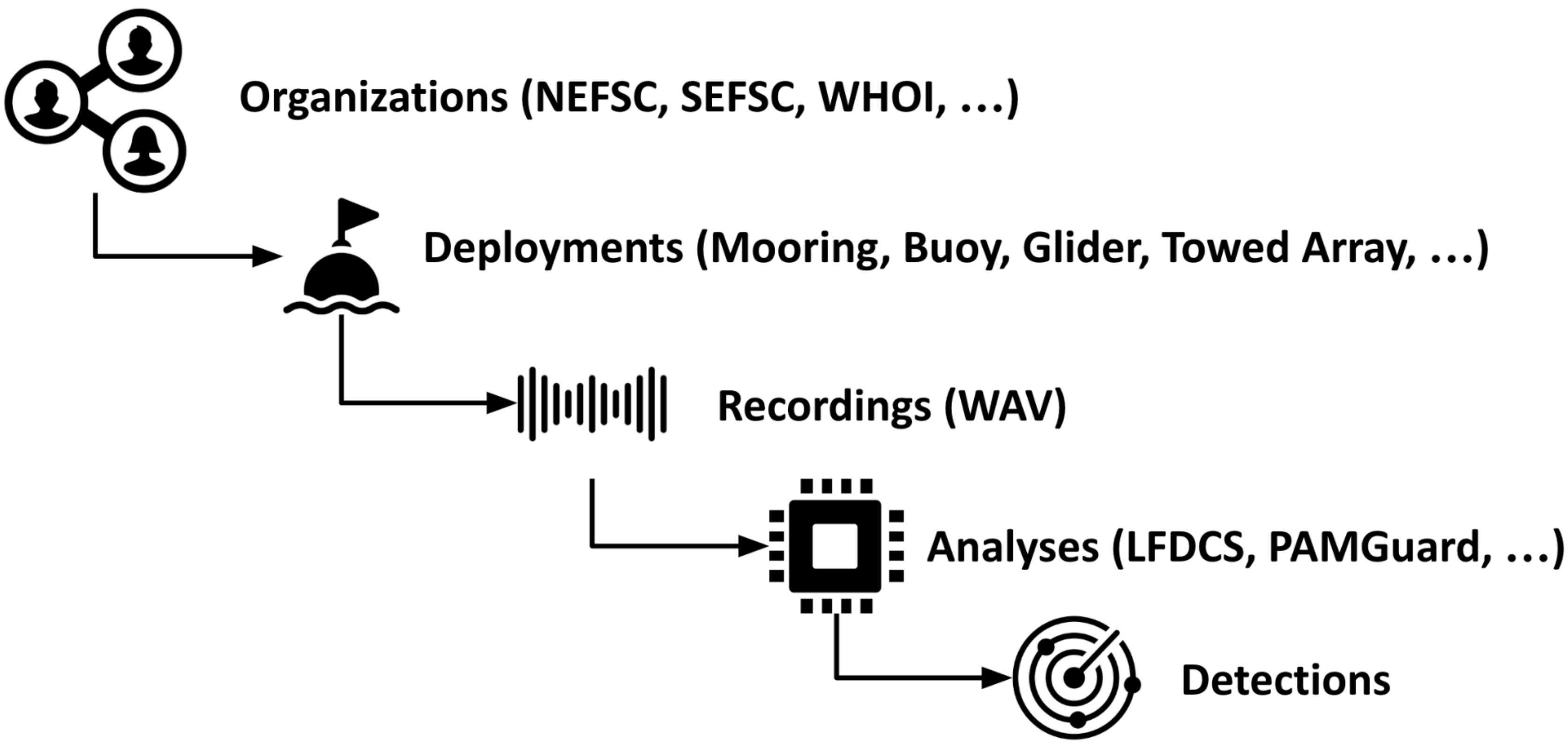

Diagram showing how the required tables are related. Each organization owns one or more deployments, for which there can be one or more recordings. The recordings are then analyzed, generating detections.

Optional Data Tables (Recommended/Required if Applicable)

The following tables are optional for any data submission as long as the desired categories already exist in the database. For each table, the data needs to be submitted only once. For example, if a device has already been submitted with a deployment, and then referenced by a deployment that is submitted later, then the Device table including that device does not need to be included in the second submission as it will already exist in Makara.

Data Table	Definition	If/When to Submit
Devices	A specific piece of equipment used as part of a deployment (e.g., platforms, hydrophones, recorders, sensors, acoustic releases, gliders, arrays)	If there are new devices that are not already in Makara
Projects	A group of deployments	If multiple deployments are submitted that the data contributor would like to have grouped together, or if new deployments are submitted that should be grouped with deployments that are already in Makara. If a project was already created with previously submitted deployment(s) and there are new deployments to be added to that project, then another Project table does not need to be submitted.
Sites	The fixed location(s) where recordings were made over the course of one or more deployments, or general region for mobile deployments	If there are new deployment sites that are not already in Makara
Recording Intervals	The range of time and/or frequencies with compromised or invalid audio (this table is not for tracking duty cycled data)	If a deployment has periods of compromised data (duty cycling is accounted for in the Recordings table, *not* the Recording Intervals table)
Tracks	The position, speed, and other variables associated with a mobile platform	If a mobile deployment is submitted
Track Positions	The instantaneous timestamped location, speed, etc. along the track of a mobile platform	If a mobile deployment is submitted

Reference Tables Maintained by NEFSC

NEFSC maintains the reference tables that are listed below. An updated copy of each table can be accessed by downloading the Excel document on the Passive Acoustic Reporting System Templates webpage. **If a data contributor is submitting data, but does not see the correct or applicable category that they need in one of these tables, they can contact the Makara Database Support Team to request that the category be added.** For example, if the submission includes detection data from an acoustic detector that does not already exist on the Detectors table in Makara, then the data contributor can contact the support team and have the new detector added to the reference table.

Reference Table	Definition
Analysis Granularity Types	The granularity of an analysis (e.g., time-binned intervals, call-by-call events, encounter events)
Analysis Processing Types	The processing method of an analysis (e.g., real-time, post-processed)
Analysis Quality Types	The analysis quality indicating the level of detection validation (e.g., full, partially, or not validated)
Behaviors	The behavior associated with a detection event (e.g., foraging, socializing)
Call Types	The unique call/click types associated with one or more sound sources/species (e.g., sperm whale clicks, North Atlantic right whale upcalls)
Call Type / Sound Sources	A list of call type codes and sound source codes indicating which call types can be used for which sound sources. This table can be used to find the call types available for a given species by filtering for that species sound_source_code.
Demographics	The demographics associated with a detection event (e.g., adult male, female or juvenile)
Detection Quality Types	The quality of detection type indicating whether it was validated or not (e.g., unverified, valid)
Detection Result Types	The detection result types based on the analysis methods (e.g., detected, possibly detected, not detected)
Detectors	The acoustic detector software programs used to process raw acoustic data (e.g., LFDCS, PAMGuard's whistle and moan detector)
Device Types	The general types of devices used as part of a deployment (e.g., platforms, hydrophones, recorders, sensors, acoustic releases, gliders, arrays)

Localization Methods	The methods used to localize the position of individual sound sources (e.g., DMON real-time tracking, time difference of arrival)
Localization Depth Methods	The methods used to localize the depth of individual sound sources (e.g., 3D Simplex, automatic direct reflected time difference of arrival)
Organizations	Any organizations that are registered in Makara (e.g., NEFSC, Woods Hole Oceanographic Institution, Cornell University)
Platform Types	The types of platforms upon which the hydrophones are mounted (e.g., gliders, moorings)
Recording Quality Types	The recording quality types indicate whether a recording is good, compromised, unusable, etc.
Sound Sources	Specific biological, anthropogenic, environmental, or unknown sound sources (e.g., North Atlantic right whale, recreational vessel, rain)

Data Type Formatting Instructions

Some fields on the data table templates require specific formatting when entering values.

Data Type	Required Formatting	Examples
Boolean	TRUE or FALSE	TRUE
Code	Uppercase letters, numbers, underscores (_), and dashes (-) only (no spaces, lowercase letters, or other symbols)	“NEFSC_MA-RI_202202_COX01”, “SOUNDTRAP-53423”
Code List	Comma-separated list of Codes within a single string	“SOUNDTRAP-53423,VEMCO-9867,FPOD-12355”
Date	Date as "YYYY-MM-DD"	"2025-03-26"
Float	Decimal number	72.56281
Integer	Integer number	1
JSON	Valid JavaScript Object Notation (JSON) for storing structured data objects and arrays (see JSON Instructions)	{ "KEY": "VALUE", "KEY": [1, 2, 3, ...], ... }
Text	Text string (no character limit)	"Recorder lost at sea"
Timestamp	Date and time in one of the following ISO-8601 formats (seconds and milliseconds are optional), must include timezone suffix (use 'Z' for UTC): "YYYY-MM-DDTHH:MMZ" "YYYY-MM-DDTHH:MM:SS±HH" "YYYY-MM-DDTHH:MM:SS±HHMM" "YYYY-MM-DDTHH:MM:SS.SSS±HHMM"	"2025-03-26T15:34Z" "2025-03-26T15:34:27+04" "2025-03-26T15:34:27-0500" "2025-03-26T15:34:27.921-0500"
Timezone	Timezone as UTC offset in one of the following formats: "UTC" "UTC±H" "UTC±HH" "UTC±HHMM"	"UTC" "UTC+4" "UTC+04" "UTC-0500"

URI	Valid Uniform Resource Indicator (URI) address for files in cloud storage. Must start with the protocol (e.g., “gs://” for Google Cloud buckets), followed by the bucket name, and then path to the file within the bucket.	"gs://nefsc-1/bottom_mounted/NEFSC_GOM/NEFSC_GOM_200410/NEFSC_GOM_200410_PU05"
URL	Valid Uniform Resource Locator (URL) address for web page	"https://dcs.whoi.edu/gom2412/gom2412_we14.shtml"

JavaScript Object Notation (JSON) Instructions

JSON (JavaScript Object Notation) is a lightweight data interchange format that is easy for humans to read and write and for machines to parse and generate. In Makara, several fields are designated as JSON type to store structured data that doesn't fit well into standard columns. For example, information about device calibration, channel maps, and detector settings can all be stored as JSON for loading directly into other software.

When to Use JSON Fields

The JSON fields are primarily intended to be used in the following situations:

- To store structured data that benefits from consistent data structures (e.g., calibration tables)
- When you need the data to be directly queryable within the database
- When the data can be read or processed directly by other software applications

JSON is particularly valuable for device specifications, channel mappings, calibration data, and complex configuration settings. However, if your additional information is primarily free-text notes or simple comments that don't require structured querying or programmatic access, NEFSC recommends using the standard comments field available in each table instead. The comments fields are simpler to fill out, don't require special formatting, and are better suited for human-readable annotations that don't need computational processing.

What is JSON?

JSON consists of two primary structures:

- **Objects**: Collections of key-value pairs enclosed in curly braces { }. Each key must be a string in double quotes, followed by a colon :, and then a value. *Example*: `{ "DETECTOR_THRESHOLD": 0.75 }`
- **Arrays**: Ordered lists of values enclosed in square brackets [ ]. *Example*: `[ 0.5, 0.75, 0.9 ]`

Values in JSON can be:

- **Strings**: Text in double quotes `"example"`
- **Numbers**: Integers or decimals without quotes `42` or `3.14`
- **Booleans**: `true` or `false` (lowercase, without quotes)
- **null**: Represented as `null` (without quotes)

There are no specific formats required for dates and timestamps. However, NEFSC recommends using the same formats listed in the Data Type Formatting Instructions table based on the ISO8601 standard (e.g., `{ "date": "2025-03-20", "datetime": "2025-03-26T15:34:27Z" }`).

Example JSON Structures

Simple object for the sensitivity of a recorder:

```
{ "SENSITIVITY": -1.8 }
```

Nested object for the high and low gain sensitivities of a recorder:

```
{ "GAIN": { "HIGH": 177.4, "LOW": 189.9 } }
```

Object with nested array for the sensitivity calibration of a hydrophone:

```
{ "SENSITIVITIES": [-170.1, -169.8, -170.2], "CALIBRATION_DATE": "2023-05-15"}
```

Complex nested structure containing a calibration frequency table represented as an array of objects where each object within the array is a row in the table:

```
{
  "CALIBRATION": [
    { "FREQ_HZ": 1, "SENSITIVITY": -170.5 },
    { "FREQ_HZ": 2, "SENSITIVITY": -171.2 },
    { "FREQ_HZ": 5, "SENSITIVITY": -172.0 },
    { "FREQ_HZ": 10, "SENSITIVITY": -172.2 },
    etc...
  ]
```

}

Recommended Guidelines

NEFSC recommends adhering to the following guidelines when creating JSON data for submission to Makara. Please note these are only guidelines, not requirements. The only requirement for these fields is that the data is in valid JSON format.

- Each entry in a JSON field should be an object, not an array or individual value so that all values have a key specifying what they are (e.g., use "`{ "SENSITIVITIES": [-170.1, -169.8, -170.2] }`" instead of just the array of values "`[-170.1, -169.8, -170.2]`"
- Always use uppercase letters for object keys to avoid changes in case across rows (e.g., `{ "HIGH": 177.4 }` and `{ "high": 177.4 }` are not compatible)

Tips for Working with JSON in Spreadsheets

- Create your JSON in a text editor or JSON validator first to ensure proper formatting
- For complex JSON structures, consider using a JSON validation tool (like jsonlint.com) before pasting into your spreadsheet
- Keep JSON structures as simple as possible while still capturing the necessary data

Consistency is Critical

It is the responsibility of the data contributor to adhere to consistent structures (aka schema) within the JSON field of each table. This is critical for ensuring data integrity and usability, and so JSON data can be readily combined across multiple rows in a given table. While NEFSC provides some example formats for each JSON field in this guide, each data contributor can develop their own JSON schema that aligns with their software and workflow.

Data Table Field Definitions

The following tables contain definitions for each field or column in the data table templates contained in the zipped folder. Each of the following field definitions tables in this document has a number of columns which are defined below in the Column Header Key section, including columns for specifying which fields are required, the type of data that should be entered (e.g., text, integer, URL), and which reference tables to refer to when filling out that field.

Column Header Key

- **Field:** The name of the field
- **Definition:** The definition of the field with examples of formatting and what to include
- **Required:** Indicates whether the field is required (i.e., 'Always') or conditional (e.g., "If recording_device_lost=FALSE, then this field is always required"). Some fields are required if the data are also being submitted to the National Centers for Environmental Information (NCEI). If those fields are not already required for submitting data to the Makara database, then there will be a note in the Required column saying "Required for NCEI".
- **Default:** Indicates whether the field has a default value if no other value or text is entered
- **Data Type:** The data type accepted by that field (see Data Type Formatting Instructions)
- **Reference Table:** The reference table that contains the category options for each field. Some reference tables are labeled with "(see Excel)", indicating that they are global reference tables available in the "Makara - Reference Tables" Excel document* in the zipped folder. If a reference table is listed here but not found with the rest of the reference tables, it is likely a reference table that is maintained internally by NEFSC and the categories can be searched for directly in Makara.
- **Constraint:** Any constraints on the contents of that field

**Note: An updated copy of each reference table can be accessed by downloading the Excel document on the Passive Acoustic Reporting System Templates webpage. If a data contributor is submitting data, but does not see the correct or applicable category that they need in one of these tables, they can contact the Makara Database Support Team to request that the category be added.*

Deployments Table Field Definitions

Below are the field definitions for the Deployments table, which is a required table.

Field	Definition	Required	Default	Data Type	Reference Table	Constraint
organization_code	Code for the organization that owns the deployment *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
deployment_code	Unique code for the deployment. The recommendation is to use [organization_code]_[project_region_code]_[YYYYMM]_[site_code]. *Examples: "WHOI_GOM_202310_WE03", "WHOI_MA-RI_202210_WE16", "NEFSC_MA-RI_202110_NS01", "NEFSC_MA-RI_202202_COX02"*	Always		Code		Unique within organization
project_code	Code of the project that the deployment belongs to. Only need to create a project code if there are multiple deployments that should be grouped together. *Examples: "WHOI_GOM_202310", "NEFSC_MA-RI_202202"*	Required for NCEI		Code	Projects	
site_code	Code of the site that the deployment was located at. For mobile platforms (e.g., gliders, drifting buoys), the site can be a general region and does not	Required for NCEI if platform is stationary		Code	Sites	

	need a unique latitude/longitude. *Examples: "COX", "GOM", "COX01", "NS02"*					
deployment_device_codes	Codes for any devices (e.g., recorders, hydrophones, platforms, releases, temperature sensors, etc.) included in the deployment. Note that recording devices from the recording_device_codes field on the Recordings table will be automatically added to this field. *Examples: "DMON2-GENERIC,SLOCUM-WE16,VEMCO-VR2W-106548", "SOUNDTRAP-6075,TEMPERATURE_SENSOR-GENERIC,VEMCO-548738"*			Code List	Devices	
deployment_platform_type_code	Type of platform *Examples: "ELECTRIC_GLIDER", "BOTTOM_MOUNTED_MOORING"*	Always		Code	Platform Types (see Excel)	
deployment_water_depth_m	Total water depth from surface to bottom (in meters) at deployment location. Value must be a positive number. *Examples: "36", "27.5"*			Float		
deployment_datetime	Date and time at start of deployment *Example: "2025-03-26T15:34:27-0500"*	Always		Timestamp		

deployment_latitude	Latitude of deployment in decimal degrees (DD.DDDDDD). If adding a mobile platform, enter the first latitude from the track. *Example: "41.141289"*	Always		Float		
deployment_longitude	Longitude of deployment in decimal degrees (DD.DDDDDD), negative for locations west of the central meridian. If adding a mobile platform, enter the first longitude from the track. *Example: "-71.102908"*	Always		Float		
deployment_vessel	Name of vessel used to deploy equipment *Example: "RV Gloria Michelle"*			Text		
deployment_cruise	Name of cruise for deployment, if applicable *Examples: "NEFSC_NE-OFFSHORE_202503_GM2503", "AMAPPS"*			Text		
deployment_alias	Alternate name for project or deployment, more applicable to historical/archived data or from collaborators *Examples: "WHOI_GOM_202310_glider03", "NEFSC_MA-RI_Feb2022_CoxLedge02"*			Text		
deployment_json	Additional structured data about deployment			JSON		

deployment_comments	Comments about deploying the equipment *Examples: "Rough weather", "5 boats within 2 miles of deployment location"*			Text		
recovery_datetime	Date and time of platform recovery *Example: "2025-09-17T07:54:28-0500"*	Required for NCEI		Timestamp		
recovery_latitude	Latitude at time of recovery in decimal degrees (DD.DDDDDD) *Example: "41.141289"*	Required for NCEI		Float		
recovery_longitude	Longitude at time of recovery in decimal degrees (DD.DDDDDD), negative for locations west of the central meridian *Example: "-71.102908"*	Required for NCEI		Float		
recovery_vessel	Name of the vessel used to recover equipment *Example: "RV Connecticut"*			Text		
recovery_cruise	Name of the cruise used for recovery, if applicable *Examples: "NEFSC_NE-OFFSHORE_202509_CT2509", "AMAPPS"*			Text		
recovery_surface_datetime	Date and time that the equipment surfaced *Example: "2025-09-17T07:51:32-0500"*			Timestamp		

recovery_burn_datetime	Date and time of signal sent to bottom-mounted recorders to corrode burn wire or activate acoustic release *Example: "2025-09-17T07:48:29-0500"*			Times tamp		
recovery_json	Additional structured data about recovery			JSON		
recovery_comments	Comments about the retrieval/recovery of equipment *Example: "Significant biofouling on hydrophone upon retrieval"*			Text		
parent_deployment_code	Deployment code of the meta-deployment the deployment belongs to (if available). Both the parent and child deployments must belong to the same organization. *Examples: "WHOI_GOM_202310_WE03", "NEFSC_MA-RI_202202_COX02", "NEFSC_HB1603"*			Code	Deployments	

Recordings Table Field Definitions

Below are the field definitions for the Recordings table, which is a required table.

Field	Definition	Required	Default	Data Type	Reference Table	Constraint
organization_code	Code of the organization that owns the recording *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
deployment_code	Unique code for the recording's deployment. *Examples: "WHOI_GOM_202310_WE03", "WHOI_MA-RI_202210_WE16", "NEFSC_MA-RI_202110_NS01", "NEFSC_MA-RI_202202_COX02"*	Always		Code	Deployments	
recording_code	Unique code to identify the recording. The recommendation is to combine the general device type with the suffix "_RECORDING". May contain additional information if there are multiple recordings from the same type of device within a deployment (e.g., recorders that have different sampling rates). Must be unique within deployment. *Examples: "SOUNDTRAP_RECORDING", "FPOD_RECORDING", "SOUNDTRAP-192KHZ_RECORDING",*	Always		Code		Unique within organization and deployment

	"SOUNDTRAP-48KHZ_RECORDING"					
recording_device_codes	Devices used to record the recording (e.g., recorder and hydrophone). At a minimum, the recorder device. Note that the device will be automatically added to the deployment_device_codes field on the Deployments table for the associated deployment. *Examples:* *"SOUNDTRAP-6078",* *"DMON2-GENERIC",* *"RESON-1951",* *"APC-1956,APC-1957,APC-1958,APC-1959,HTI-1943,HTI-1971"*	Always		Code List	Devices	
recording_start_datetime	Date and time of the start of the recording *Example:* *"2025-03-26T15:34:27-0500"*	If recording_device_lost=FALSE, then this field is always required		Timestamp		
recording_end_datetime	Date and time of the end of the recording *Example:* *"2025-09-17T07:54:28-0500"*	Required for NCEI		Timestamp		
recording_duration_secs	The duration of recording per interval (the amount of time, in seconds, the recorder is turned "on" within the recording cycle). For continuous recordings, the recording_duration_secs and recording_interval_secs will both be equal to the number of seconds in a file, or a set	If recording_device_lost=FALSE, then this field is always required		Float		

	amount of time if sound file lengths varied for continuous recording. *Examples: If the first 10 minutes of every hour is recorded, then recording_duration_secs is "600" and recording_interval_secs is "3000". For continuous recordings, both fields might be set to 1 hour for both, or "3600".*					
recording_interval_secs	The interval over which a duty cycled recording takes place (the total amount of time, in seconds, the recorder is turned "on" and "off" within the recording cycle). For continuous recordings, this entry will be equal to the recording_duration_secs. *Examples: If the first 10 minutes of every hour is recorded, then recording_duration_secs is "600" and recording_interval_secs is "3000". For continuous recordings, both fields might be set to 1 hour for both, or "3600".*	If recording_device_lost=FALSE, then this field is always required		Float		
recording_sample_rate_khz	Sample rate of recording in kHz *Examples: "2", "48, "96", "34.5"*	If recording_device_lost=FALSE, then this field is always required		Float		
recording_bit_depth	Resolution in bit depth of recording	Required for NCEI		Integer		

	Examples: "12", "16", "24"					
recording_channel	The acoustic data channel assigned. For single channel recordings, this value will be 1. For multichannel recordings, the value should be the corresponding channel number in the recordings. *Examples: For single channel recordings, the value will be "1". For multichannel recordings, the value might be "2", "5", "12", etc. depending on which channel is selected for the submitted recording (if a particular one was selected).*	Required for NCEI		Intege r		
recording_n_channels	The total number of channels in the recording file. This will be 1 for a single channel file and greater than 1 for a multichannel file. *Examples: For single channel recordings, the value will be "1". For multichannel recordings, the value might be "2", "5", "12", etc. depending on how many channels were recorded total.*	Always		Intege r		
recording_filetypes	Soundfile type extension(s). If multiple types exist, list all. *Examples: "WAV", "AIF"*			Text		
recording_timezone	Timezone that the soundfiles are recorded in and correspond to, in relation to UTC. For deployments that are spread across timezone changes and sound files have two different	Always		Timez one		

	timezones (beginning and end of recordings have different timezones), they should each be entered as their own recording ID for that timezone. *Examples: "UTC", "UTC-5", "UTC-4", "UTC+8"*					
recording_usable_start_datetime	Date and time of the beginning of usable data; if all data is usable, this should be the same as the start time of the first viable recording in the water. *Example: "2025-03-26T15:34:27-0500"*			Timestamp		
recording_usable_end_datetime	Date and time of the end of usable data; if all data is usable, this should be the same as the end time of the last viable recording in the water before retrieved or acoustic release signal sent. *Example: "2025-09-17T07:54:28-0500"*			Timestamp		
recording_usable_min_frequency_khz	The minimum usable frequency (in kHz) of the sound file *Examples: "1", "4.5"*			Float		
recording_usable_max_frequency_khz	The maximum usable frequency (in kHz) of the sound file *Examples: "36", "98.5"*			Float		
recording_quality_code	Quality type of the entire recording. Defaults to "UNVERIFIED" meaning recording has not been quality reviewed. Must be	Required for NCEI	UNVERIFIED	Code	Recording Quality Types (see Excel)	

	"COMPROMISED" if the recording has any recording_intervals that are "COMPROMISED" or "UNUSABLE". *Examples: "COMPROMISED", "GOOD", "UNUSABLE", "UNVERIFIED"*					
recording_device_lost	Indicates whether the recorder was not retrieved. Defaults to "FALSE", i.e., the recorder was retrieved. *Examples: "TRUE", "FALSE"*		FALSE	Boolean		
recording_device_depth_m	Depth of the hydrophone (in meters) at deployment location *Examples: "3", "37.1"*	Required for NCEI		Float		
recording_redacted	Indicates if data has been redacted by the Navy in which some or all of data is not publicly available *Examples: "TRUE", "FALSE"*			Boolean		
recording_json	User-adjusted gain on recording device, i.e., (in dB) for array and drop hydrophone systems or (High or Low) for Soundtraps	Recording-specific calibration required for NCEI		JSON		
recording_uri	The Uniform Resource Identifier (URI) for the audio recordings, if available. For Google Cloud Buckets, URIs should be in the form “gs://<bucket>/<key>”. *Example: If your recording is stored on a Google bucket, the URI pathway might be*			URI		

	something like "gs://nefsc-1/bottom_mounted/NEFSC_GOM/NEFSC_GOM_200410/NEFSC_GOM_200410_PU05"					
recording_comments	Comments about the recording *Examples: "Hydrophone was out of the water for the first 30 minutes of recording"*			Text		

Analyses Table Field Definitions

Below are the field definitions for the Analyses table, which is a required table.

Field	Definition	Required	Default	Data Type	Reference Table	Constraint
deployment_organization_code	Code of the organization that the deployment belongs to *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
deployment_code	Unique code for the deployment that the analysis was performed on. *Examples: "WHOI_GOM_202310_WE03", "WHOI_MA-RI_202210_WE16", "NEFSC_MA-RI_202110_NS01", "NEFSC_MA-RI_202202_COX02"*	Always		Code	Deployments	
recording_codes	Unique code(s) to identify the recording(s) belonging to the deployment that were used for the analysis. *Examples: "SOUNDTRAP_RECORDING", "FPOD_RECORDING", "SOUNDTRAP-192KHZ_RECORDING", "SOUNDTRAP-48KHZ_RECORDING"*	Always		Code List	Recordings	
analysis_organization_code	Code of the organization that conducted and owns the analysis. *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	

analysis_code	Unique code for the analysis, generally contains the individual or group of species in all caps followed by "_ANALYSIS". Must be unique within the associated deployment. *Examples: "RIWH_ANALYSIS", "BEAKED_ANALYSIS"*	Always		Code		Unique within analysis organization and deployment
analysis_sound_source_codes	The species or detected sound(s) the analysis was conducted for. Multiple entries separated by commas allowed (i.e., an analysis may have looked for both MIWH and HUWH simultaneously). *Examples: "HUWH", "STDO", "RV-L", "MIWH,HUWH,RIWH,SEWH,FIWH"*	Always		Code List	Sound Sources (see Excel)	
analysis_granularity_code	Granularity type at which the analysis was conducted (i.e., time-binned versus event-based) *Examples: "EVENT_CALL", "EVENT_ENCOUNTER", "INTERVAL"*	Always		Code	Analysis Granularity Types (see Excel)	
analysis_start_datetime	Start date and time for detection analysis (i.e., the start date and time for when the data were analyzed by human analyst or detector) *Example: "2025-03-26T15:34:27-0500"*	Required for NCEI		Timestamp		
analysis_end_datetime	End date and time for detection analysis (i.e., the end date and	Required for NCEI		Timestamp		

	time for when the data were analyzed by human analyst or detector) *Example: "2025-09-17T07:54:28-0500"*					
analysis_sample_rate_khz	Sample rate (in kHz) of acoustic data used for detector and analysis, this may be a resampled rate than the original recordings *Examples: "1", "10.5"*	Always		Float		
analysis_min_frequency_khz	The minimum frequency (in Hz) used for the analysis, if applicable. The default value is "0" (Hz). *Example: If middle frequencies were viewed for the analysis (e.g., 200-1000Hz), this value would be "200".*	Required for NCEI	0	Float		
analysis_max_frequency_khz	The maximum frequency (in Hz) used for the analysis, if applicable. The value is relative to the recording's sample rate. *Examples: If the original recordings had a sample rate of 48kHz, and the data was either resampled to 2kHz for the analysis, or only viewed at up to 1kHz, then this value would be "1000". The value for this field would be "24000" if the sampling rate was kept at 48kHz.*	Required for NCEI		Float		
analysis_processing_code	Processing type of the analysis	Always		Code	Analysis Processing	

	Examples: "POST_PROCESSED", "REAL_TIME"				Types (see Excel)	
analysis_quality_code	Analysis quality type indicating the level of detection validation at the level of the analysis granularity *Examples: "FULLY_VALIDATED", "NOT_VALIDATED", "PARTIALLY_VALIDATED"*	Always		Code	Analysis Quality Types (see Excel)	
analysis_protocol_reference	Citation with URL and/or DOI (if available) for published paper for detector and the document with the analysis methods used for validating the data (published protocols are preferred, but can refer to a name of an internal document that is not currently published). Separate multiple entries with semicolons. *Examples: "NEFSC Whale Presence Analysis Protocol; Baumgartner and Mussoline 2011, doi:10.1121/1.3562166; FPOD manual 2.0"*	Always		Text		
analysis_dataset_url	Published Uniform Resource Locator (URL) for dataset, if available *Example: "https://dcs.whoi.edu/gom2412/gom2412_we14.shtml"*			URL		
analysis_release_data	Indicates whether the analysis can be released by NEFSC Passive Acoustics Branch on	Always		Boolean		

	request via North Atlantic Right Whale Consortium database *Examples: "TRUE", "FALSE"*					
analysis_release_pacm	Indicates whether the analysis can be shown on PACM *Examples: "TRUE", "FALSE"*	Always		Boolean		
detector_codes	The detector(s) used for the analysis *Examples: "LFDCS", "MANUAL", "ISRAT_COD", "PAMGUARD_WHISTLE_MOAN,PAMGUARD_CLICK,", "MIWH_DNN"*	Always		Code List	Detectors (see Excel)	
detector_version	The version of the detector used for the analysis *Examples: "2.0", "beta"*			Text		
detector_json	Detector settings in JSON format			JSON		
detector_output_filename	Name of detector output file *Example: "NEFSC_MA-RI_202309_NS12_RIWH_DAILY_UTC-5"*			Text		
detector_output_uri	The Uniform Resource Identifier (URI) for the detector output, if available. For Google Cloud Buckets, URIs should be in the form "gs://<bucket>/<key>" *Example: If your detector output is stored on a Google bucket, the URI pathway might be something like "gs://nefsc-1-detector-output/LFDCS_BALEEN/NEFSC_GOM/N*			URI		

	EFSC_GOM_200410/NEFSC_GOM_200410_PU05"					
analysis_analysts	The analyst(s) who conducted the analysis. List all applicable. The recommendation is to use all capitals with the first initial then last name. If multiple analysts worked on the same deployment, they can all be included with comma separation. *Examples: "JWILDER", "GDAVIS", "JWILDER,GDAVIS"*			Text		
analysis_json	Any collection of settings that specify how the analysis or methods were done, must be in valid JSON format *Example:* `{ "MAX_MAHALANOBIS_DISTANCE": 3 }`			JSON		
analysis_comments	Comments about the analysis *Example: "Data were analyzed by Julianne Wilder up to 2025-03-26T15:34:30-0500 and then the remaining data from the deployment were analyzed by Genevieve Davis"*			Text		

Detections Table Field Definitions

Below are the field definitions for the Detections table, which is a required table.

Field	Definition	Required	Default	Data Type	Reference Table	Constraint
deployment_organization_code	Code of the organization that the deployment belongs to. *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
deployment_code	Unique code for the deployment that the analysis belongs to. *Examples: "WHOI_GOM_202310_WE03", "WHOI_MA-RI_202210_WE16", "NEFSC_MA-RI_202110_NS01", "NEFSC_MA-RI_202202_COX02"*	Always		Code	Deployments	
analysis_organization_code	Code of the organization that conducted and owns the analysis. *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
analysis_code	Code of the analysis that the detection belongs to. *Examples: "RIWH_ANALYSIS", "BEAKED_ANALYSIS"*	Always		Code	Analyses	
detection_start_datetime	Start date and time for the time period for the detection event or time bin depending on analysis granularity *Example: "2025-03-26T15:34:27-0500"*	Always		Timestamp		

detection_end_datetime	End date and time for the time period for the detection event or time bin depending on analysis granularity. If detection is for a single point in time, the start and end datetimes can be the same. *Examples: "2025-03-26T15:34:27-0500", "2025-03-26T15:34:30-0500"*	Always		Timestamp		
detection_effort_secs	The amount of time (in seconds) the effort occurred in. This would translate to encounter or event duration for encounter/event level data. If the entire period was analyzed, enter total duration between the detection start and end datetimes. *Example: If the first 5 minutes of every hourly analysis bin was analyzed, this number would be "300". If the full hour of every hourly analysis bin was analyzed, this number would be "3600".*	Always		Float		
detection_sound_source_code	The species or detected sound the detection period relates to *Examples: "HUWH", "STDO", "RV-L", "RAIN"*	Always		Code	Sound Sources (see Excel)	
detection_call_type_code	The call/click type used for presence determination for the detection period	Always		Code	Call Types (see Excel)	

	Examples: "RIWH_UPCALL", "HUWH_SONG", "OD_WHIS_HF"					
detection_behavior_code	Behavior associated with detected event *Examples: "CONTACT", "FORAGING", "MOM_CALF", "MATING"*			Code	Behaviors (see Excel)	
detection_demographic_code	The demographic determined of the sound source for the detection period *Examples: "ADULT_FEMALE_OR_JUV", "MID-SIZED_INDIVIDUAL", "MIXED_GROUP"*			Code	Demographics (see Excel)	
detection_n_validated	Number of actual detections that have been validated to be correct in the detection time period *Examples: "3", "258"*			Integer		
detection_n_total	Total number of detections for the sound source within the detection time period (can include true and false detections) *Examples: "15", "179"*			Integer		
detection_result_code	Detection result indicating if the species was detected or not *Examples: "DETECTED", "POSSIBLY_DETECTED", "NOT_DETECTED"*	Always		Code	Detection Result Types (see Excel)	
detection_event_type	Type of detection event (if available)			Text		

	Example: PAMGuard's eventType field					
detection_event_id	ID of the detection event (if available) *Example: PAMGuard's event ID*			Text		
detection_latitude	Latitude of the platform location at time of detection (or start of encounter) in decimal degrees (DD.DDDD), only applicable to mobile platforms (e.g., gliders, buoys) *Example: "41.141289"*			Float		
detection_longitude	Longitude of the platform location at time of detection (or start of encounter) in decimal degrees (DD.DDDD), only applicable to mobile platforms (e.g., gliders, buoys). Negative for locations west of the central meridian. *Example: "-71.102908"*			Float		
detection_received_level_db	Received level of the detection in dB. For single call detections, a singular value should be placed here. For encounters or time binned data, an average or median received level should be placed here. Further details such as the exact unit (e.g. dB RMS, or dB pp), and whether a median or average received level was used should be placed in the analysis_json or analysis_comments fields of the parent analysis. For encounters/time binned data,			Float		

	each signal's received level can be placed in the detection_json field. *Example: "15"*					
detection_quality_code	Quality type of the detection if it was validated or not. This field may be left blank if the analysis_quality_code is "UNKNOWN" for the parent analysis, otherwise it is recommended to fill it out. *Examples: "INVALID", "UNVERIFIED", "VALID", "PARTIALLY_VALID"*			Code	Detection Quality Types (see Excel)	
detection_n_animals	Estimated number of animals thought to be in the detection *Example: "25"*			Intege r		
detection_n_animals_min	Minimum number of animals thought to be in the detection *Example: "1"*			Intege r		
detection_n_animals_max	Maximum number of animals thought to be in the detection *Example: "3"*			Intege r		
detection_json	Additional structured data about the detection			JSON		
detection_comments	Comments about the detection			Text		
localization_method_code	Method used to localize the position of the animal(s) *Examples: "3D_SIMPLEX", "DMON_REAL-TIME", "SHIP-POSITION"*	Required if submitting localization data		Code	Localization Methods (see Excel)	

localization_latitude	Estimated localized latitude of the animal in decimal degrees (DD.DDDDDD) *Example: "41.141289"*			Float		
localization_latitude_min	Minimum localized latitude accounting for localization uncertainty in decimal degrees (DD.DDDDDD) *Example: "40.724307"*			Float		
localization_latitude_max	Maximum localized latitude accounting for localization uncertainty in decimal degrees (DD.DDDDDD) *Example: "41.141289"*			Float		
localization_longitude	Estimates localized longitude of the animal in decimal degrees (DD.DDDDDD), negative for locations west of the central meridian *Example: "-71.102908"*			Float		
localization_longitude_min	Minimum localized longitude accounting for localization uncertainty in decimal degrees (DD.DDDDDD), negative for locations west of the central meridian *Example: "-70.016567"*			Float		
localization_longitude_max	Maximum localized longitude accounting for localization uncertainty in decimal degrees (DD.DDDDDD), negative for locations west of the central meridian			Float		

	Example: "-71.102908"					
localization_distance_m	Estimated range of detection (in meters) from the recording platform *Examples: "3", "60.5"*			Float		
localization_distance_m_min	Minimum range of detection (in meters) from the recording platform *Example: "0"*			Float		
localization_distance_m_max	Maximum range of detection (in meters) from the recording platform *Example: "8047"*			Float		
localization_bearing	For beamforming, the estimated bearing angle of the localization cone (0-360 degrees). Specify if bearing is measured relative to true north, magnetic north, or device orientation, and compass rose convention (clockwise, counter-clockwise) in analysis_json or analysis_comments. *Examples: "170", "170.5"*			Float		
localization_bearing_min	For beamforming, the start bearing angle of the localization cone (0-360 degrees) *Examples: "167", "167.5"*			Float		
localization_bearing_max	For beamforming, the end bearing angle of the localization cone (0-360 degrees) *Example: "175", "175.34"*			Float		

localization_depth_method_code	Method used to estimate depth of the animal(s) *Examples: "3D_TDOA", "MANUAL_DRTD", "WHALEDO"*			Code	Localization Depth Methods (see Excel)	
localization_depth_n_signals	Number of signals used to derive the depth *Examples: "7", "218"*			Intege r		
localization_depth_m	Estimated depth of animal (in meters) *Examples: "5", "28.6"*			Float		
localization_depth_m_min	Minimum depth of animal (in meters) *Example: "1.5"*			Float		
localization_depth_m_max	Maximum depth of animal (in meters) *Example: "893"*			Float		
localization_json	Additional structured data about localization in JSON format			JSON		
localization_comments	Comments about the localization			Text		

Devices Table Field Definitions

Below are the field definitions for the Devices table, which is an optional table. It is only required if one or more new devices need to be added to Makara.

Field	Definition	Required if Submitted	Default	Data Type	Reference Table	Constraint
organization_code	Code of the organization that owns the device *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
device_code	Unique code to identify the device. The recommendation is to combine the device type with the serial number as the suffix, i.e., [DEVICE MODEL]-[DEVICE SERIAL NUMBER]. If there is no serial number and/or the device is not being tracked individually, the serial number can be replaced with the suffix "-GENERIC". *Examples: "SOUNDTRAP-6075", "VEMCO-548738", "VEMCO-VR2W-106548", "TEMPERATURE_SENSOR-GENERIC", "DMON2-GENERIC", "SLOCUM-WE16"*	Always		Code		Unique within organization
device_type_code	The code for the type of device *Examples: "RECORDING_DEVICE", "TEMPERATURE_SENSOR", "ACOUSTIC_RELEASE", "ELECTRIC_GLIDER", "ACTIVE_RECEIVER"*	Always		Code	Device Types (see Excel)	

device_manufacturer	The name of the manufacturer that produced the device *Examples: "Ocean Instruments", "Vemco", "Woods Hole Oceanographic Institution", "Teledyne Webb Research"*			Text		
device_model_name	The name of the particular model of the device *Examples: "STD", "VR2AR", "DMON", "Slocum glider"*			Text		
device_model_number	The model number, ID, or version of the device as issued by the manufacturer, if applicable *Examples: "ST600 STD", "2", "3.0"*	Required for NCEI		Text		
device_serial_number	The serial number of the device as assigned by the manufacturer *Examples: "6075", "106548"*			Text		
device_description	A more detailed description of the device and/or the long-form or common name *Examples: "SoundTrap600", "Temperature sensor", "VR2AR Acoustic release and receiver", "Collaborator DMON", "Collaborator Slocum glider we03", "Collaborator VR2W Acoustic receiver", "Moana"*			Text		
device_start_date	The start date when the device was acquired or began being used actively			Date		

	Example: "2019-09-25"					
device_end_date	The end date when the device was decommissioned, lost, or is no longer active *Example: "2021-12-03"*			Date		
device_alias	Any alternate or previous names, nicknames, or identifiers for the device *Example: If the device is a Marine Autonomous Recording Unit (MARU) that is currently identified as "MARU-59863" but used to be called "POPUP-59863", then "POPUP-59863" can go in this field.*			Text		
device_json	Additional structured data about the device *Example:* `{ "GAIN_HIGH": 177.4, "GAIN_LOW": 189.9 }`	Device-specific calibration required for NCEI		JSON		
device_comments	Comments about the device *Examples: "designated for PA group project", "TAG_ID=A69-1601-61202", "CD_NUMBER=CD0004359172; designated for ONMS"*			Text		
child_device_codes	Comma-separated list of unique codes to identify multiple child devices that are associated with a parent device (see Example #3 for a full example of this for a towed array)			Code List	Devices	

	Example: If a towed array of hydrophones is submitted, the parent device code might be "LINEAR_TOWED_ARRAY-3", and the child device codes for the individual hydrophones and other devices associated with the array might be "HYDROPHONE_INDIVIDUAL-1,HYDROPHONE_INDIVIDUAL-2,HYDROPHONE_INDIVIDUAL-3,DEPTH_SENSOR-2740,TEMPERATURE_SENSOR-GENERIC"					

Projects Table Field Definitions

Below are the field definitions for the Projects table, which is an optional table. It is only required if multiple deployments are submitted that the data contributor would like to have grouped together, or if new deployments are submitted that should be grouped with deployments that are already in Makara.

Field	Definition	Required if Submitted	Default	Data Type	Reference Table	Constraint
organization_code	Code of the organization that owns or oversees the project *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
project_code	Unique code to identify the project. NEFSC uses the recommendation [organization_code]_[project_region_code]_[YYYYMM] where the year/month indicates the start of the project, but the naming conventions can be different for other organizations as necessary. *Examples: "WHOI_GOM_202310", "NEFSC_MA-RI_202202"*	Always		Code		Unique within organization
project_name	The name of the project, which can be set to the project_code or refer to another naming convention (usually applies to historical data and/or towed array deployments) *Examples: "SanctSound", "Atlantic Cod Project", "WHOI_MA-RI_202202"*	Required for NCEI		Text		
project_description	A more detailed narrative description of the project	Required for NCEI		Text		

	Examples: "Acoustic data collection in near real time that is uploaded to the Robots4Whales website (https://robots4whales.whoi.edu/) in collaboration with Mark Baumgartner at WHOI.", "Cox Ledge and Nantucket Shoals deployments; primary monitoring goals for NARWs on Nantucket Shoals and Cod Spawning grounds on Cox Ledge; then long term monitoring in Southern New England"					
project_alias	Any alternate or previous names, nicknames, or identifiers for the device, more applicable to historical/archived data or from collaborators *Examples: "SanctSound", "Atlantic Cod Project"*			Text		
project_contacts	The primary person(s) responsible for the project, i.e., Person(s) of Contact, generally including their name and email address *Example: "Sofie Van Parijs <sofie.vanparijs@noaa.gov>"*	Always		Text		
project_funding	Describe the funding source for this project *Examples: "Navy Program N45", "NSF Grant #905302"*			Text		
project_collaborator_organization_codes	Comma-separated codes for any other organizations that			Code List	Organizations (see Excel)	

	have collaborated on the project (not including the primary organization overseeing the project) *Examples: "NEFSC", "MADMF,TNC,UMASSDART,GARFO"*					
project_comments	Comments about the project *Example: "Project was originally associated with the SanctSound project but has now become an independent project. Any future deployments in this region should be associated with this project."*			Text		

Sites Table Field Definitions

Below are the field definitions for the Sites table, which is an optional table. It is only required if one or more new deployment sites need to be added to Makara.

Field	Definition	Required if Submitted	Default	Data Type	Reference Table	Constraint
organization_code	Code of the organization that owns or oversees the deployment(s) at that site *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
site_code	Unique code to identify the site *Examples: "COX", "GOM", "COX01", "NS02", "USTR11"*	Always		Code		Unique within organization
site_name	The descriptive name of the site, which can be set to the site_code or refer to another naming convention, if applicable *Examples: "cox", "gom", "COX", "GOM", "COX01", "NS02"*			Text		
site_description	A more detailed narrative description of the site *Examples: "Woods Hole Oceanographic Institution's Cox Ledge real-time Slocum glider", "Cox Ledge 01", "Nantucket Shoals"*			Text		
site_latitude	Latitude of site in decimal degrees (DD.DDDDDD) for stationary platforms *Example: 41.141289*			Float		
site_longitude	Longitude of site in decimal degrees (DD.DDDDDD) for			Float		

	stationary platforms, negative for locations west of the central meridian *Example: -71.102908*					
site_alias	Any alternate or previous names, nicknames, or identifiers for the site, more applicable to historical/archived data or from collaborators *Examples: "cox-01", "Cox Ledge 001", "Nantucket Shoals site 5"*			Text		
site_comments	Comments about the site *Example: "Site was originally chosen due to its proximity to the Boston shipping lane"*			Text		

Recording Intervals Table Field Definitions

Below are the field definitions for the Recording Intervals table, which is an optional table. It is only required if one or more recordings have periods of compromised data (duty cycling is accounted for in the Recordings table, *not* the Recording Intervals table). For an example of a recording_intervals table that is filled out, see Example #1.

Field	Definition	Required if Submitted	Default	Data Type	Reference Table	Constraint
organization_code	Code of the organization that owns the recording *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
deployment_code	Unique code for the recording's deployment. *Examples: "WHOI_GOM_202310_WE03", "WHOI_MA-RI_202210_WE16", "NEFSC_MA-RI_202110_NS01", "NEFSC_MA-RI_202202_COX02"*	Always		Code	Deployments	
recording_code	Unique code to identify the recording. *Examples: "SOUNDTRAP_RECORDING", "FPOD_RECORDING", "SOUNDTRAP-192KHZ_RECORDING", "SOUNDTRAP-48KHZ_RECORDING"*	Always		Code	Recordings	
recording_interval_channel	The acoustic data channel of the recording interval. For single channel recordings, this value will be 1 or can be left blank. For multichannel recordings, the value should be the	Required for NCEI		Integer		

	corresponding channel number in the recordings. *Examples: For single channel recordings, the value will be "1" or can be left blank. For multichannel recordings, the value might be "2", "5", "12", etc. depending on which channel has the recording intervals.*					
recording_interval_start_datetime	Date and time of the start of the compromised or unusable recording interval *Example: 2021-10-15T09:57:53+0000*	Required for NCEI		Timestamp		
recording_interval_end_datetime	Date and time of the end of the compromised or unusable recording interval *Example: 2021-10-18T13:57:53+0000*	Required for NCEI		Timestamp		
recording_interval_min_frequency_khz	The minimum frequency (in kHz) of the compromised or unusable recording interval, if applicable *Examples: "1", "4.5"*	Required for NCEI		Float		
recording_interval_max_frequency_khz	The maximum frequency (in kHz) of the compromised or unusable recording interval, if applicable *Examples: "1", "4.5"*	Required for NCEI		Float		
recording_interval_quality_code	Quality type of the recording interval	Always		Code	Recording Quality Types (see Excel)	

	Examples: "UNUSABLE", "COMPROMISED"					
recording_interval_json	Additional structured data about the recording interval			JSON		
recording_interval_comments	Comments about the recording interval that explain why it is compromised or unusable *Examples: "Data gap from 2021-10-15T09:57:53+0000 to 2021-10-18T13:57:53+0000 due to hard drive malfunction", "Unusable data due to daylight savings time change"*			Text		

Tracks Table Field Definitions

Below are the field definitions for the Tracks table, which is an optional table. It is only required if one or more mobile deployments are being submitted to Makara. Note that track data will be downsampled to 15-minute frequency for storage in Makara.

Field	Definition	Required if Submitted	Default	Data Type	Reference Table	Constraint
organization_code	Code for the organization that owns or oversees the mobile deployment that produced the track *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
deployment_code	Unique code to identify the track of the mobile deployment. *Examples: "WHOI_GOM_202310_WE03", "WHOI_MA-RI_202210_WE16", "NEFSC_MA-RI_202110_NS01", "NEFSC_MA-RI_202202_COX02"*	Always		Code	Deployments	
track_code	Code to identify the mobile platform track. The recommendation is to combine the platform type with the suffix "_TRACK". *Examples: "GLIDER_TRACK", "SHIP_TRACK", "TOWED-ARRAY_TRACK"*	Always		Code		Unique within organization and deployment
track_uri	The Uniform Resource Identifier (URI) for external file containing track data, if available. For Google Cloud Buckets, URIs should be in the form “gs://<bucket>/<key>”. Useful			URI		

	for linking tracks to high resolution data files.					
track_json	Additional structured data about the track			JSON		
track_comments	Comments about the track *Example: "Glider drifted west on 2024-10-27 due to storm, resumed planned course on 2024-10-28"*			Text		

Track Positions Table Field Definitions

Below are the field definitions for the Track Positions table, which is an optional table. It is only required if one or more mobile deployments are being submitted to Makara.

Field	Definition	Required if Submitted	Default	Data Type	Reference Table	Constraint
organization_code	Code for the organization that owns or oversees the mobile deployment that produced the track *Examples: "NEFSC", "WHOI"*	Always		Code	Organizations (see Excel)	
deployment_code	Unique code to identify the track of the mobile deployment. *Examples: "WHOI_GOM_202310_WE03", "WHOI_MA-RI_202210_WE16", "NEFSC_MA-RI_202110_NS01", "NEFSC_MA-RI_202202_COX02"*	Always		Code	Deployments	
track_code	Code to identify the mobile platform track for this position. The recommendation is to combine the platform type with the suffix "_TRACK". *Examples: "GLIDER_TRACK", "SHIP_TRACK", "TOWED-ARRAY_TRACK"*	Always		Code	Tracks	
track_position_datetime	Date and time of the individual track position *Example: 2023-10-12T18:09:05+0000*	Always		Timestamp		

track_position_latitude	Latitude of individual track position in decimal degrees (DD.DDDDDD) *Example: 44.4427*	Always		Float		
track_position_longitude	Longitude of individual track position in decimal degrees (DD.DDDDDD), negative for locations west of the central meridian *Example: -67.2215*	Always		Float		
track_position_speed_knots	Speed (in knots) at which the platform was moving at the time of the individual track position *Examples: "5", "8.2"*			Float		
track_position_depth_m	Water depth (in meters) of the hydrophone at the time of the individual track position *Examples: "3", "37.1"*			Float		
track_position_json	Additional structured data about track positions			JSON		
track_position_comments	Comments about the track positions *Example: "On 2025-04-17, the waypoints were changed to direct the glider south to investigate a gathering of North Atlantic right whales that had been sighted in Cape Cod Bay by the NEFSC aerial team."*			Text		

Frequently Asked Questions

Where can I find the submission templates to enter my data/metadata?

The data template tables and example tables are provided in a downloadable zipped folder on the Passive Acoustic Reporting System Templates webpage.

What happens if I find a mistake in my metadata after I have submitted my templates? For example, you submit your recordings data in local time but realize later you need to change it to UTC.

Not to worry! You have two options: 1) you can resubmit a whole table (or tables) with the corrected data, or 2) you can go to Makara directly and correct a field or fields. For example, if you change the site_code in the Sites table, all of the deployments that reference that site_code will automatically be updated to the new site_code.

Example Scenarios

The following scenarios describe potential situations that may arise when submitting data and metadata to Makara. Each scenario is associated with a set of example templates provided in the zipped folder that provide a mockup of what that scenario might look like. Within the template zipped folder is a sub-directory named "examples/". This sub-directory contains four example folders containing CSV files that are filled out according to each scenario.

Example #1: Single moorings with multiple same recorder types but different settings

Location of example templates within zipped folder: examples/EXAMPLE_1_Moorings_Same_Recorder_Type

This example shows two single, bottom-mounted moorings, each equipped with two SoundTraps, a temperature sensor, and a Vemco receiver. The NS01 mooring was deployed near Nantucket Shoals off the coast of Massachusetts and Rhode Island from October 2021 to February 2022. It had one SoundTrap600 (ST600 STD) and one high-frequency SoundTrap600 (ST600 HF) to record harbor porpoises. Both devices recorded over the whole deployment, but the ST600 HF was duty-cycled, recording for 5 minutes at the beginning of each hour with a sampling rate of 384 kHz. The ST600 STD recorded continuously with a lower sampling rate of 64 kHz. This example also shows what it looks like when there are compromised data in one of the recordings (e.g., a data gap in the recording from the ST600 HF), which can be viewed on the recording_intervals table. Daily presence of North Atlantic right whales was analyzed for 6 days from the ST600 STD recording, and hourly presence of harbor porpoises was analyzed for 6 days from the ST600 HF recording.

The COX01 mooring was deployed near Cox Ledge off the coast of Massachusetts and Rhode Island from November 2021 to February 2022. It had two ST600 HF hydrophones that recorded continuously with a sampling rate of 384 kHz, but one is only on and recording for the first half of the deployment (2021-11-03 to 2022-01-01) and the other is recording for the second half of the deployment (2022-01-01 to 2022-02-24; see the recordings table). Daily presence of North Atlantic right whales and hourly presence of harbor porpoises were analyzed for 6 days from each of the COX01 recordings. *Note: More analyses were performed over the whole deployment, but for this example we only showed 6 days worth of analysis.*

Example #2: Single moorings with multiple different recorder types

Location of example templates within zipped folder: examples/EXAMPLE_2_Moorings_Different_Recorder_Types

This example shows a single, bottom-mounted mooring that was deployed in the offshore Gulf of Maine along the northern edge of Georges Bank from July to December 2022. The mooring was equipped with two hydrophones (one SoundTrap600 with a sampling rate of 64 kHz and one FPOD with a sampling rate of 1,000 kHz to record harbor porpoise), a temperature sensor, and a Vemco receiver. Both devices recorded continuously over the whole deployment, but the FPOD recordings were automatically deleted once they were processed in real-time by a click detector onboard the platform (so only detection data were collected from the FPOD, no audio data). Daily presence of blue whales were analyzed for 6 days from the ST600 STD recording, and hourly presence of harbor porpoises were determined automatically from the detection data for 6 days from the FPOD data. *Note: More analyses were performed over the whole deployment, but for this example we only showed 6 days worth of analysis.*

Example #3: Towed array deployments across multiple cruise legs

Location of example templates within zipped folder: examples/EXAMPLE_3_Towed_Array

This example shows recordings from five towed array deployments across different legs of a research cruise that surveyed offshore of the northeast U.S. in June-August 2016. The towed array setup consisted of two separate oil-filled arrays (an inline and an endline array), each containing multiple hydrophones and one containing a depth sensor. There were different types of hydrophones (e.g., APC, Reson, HTI), all of which recorded continuously whenever the array was deployed. Some had a sampling rate of 192 kHz while others had a sampling rate of 500 kHz to record Kogia clicks.

All of the 192 kHz recordings were post-processed and analyzed for the clicks of various beaked whale species including Blainville's, goose-beaked, Gervais', unidentified True's/Gervais', Sowerby's, True's, and unidentified Mesoplodon. Three of the 500 kHz recordings from deployments NEFSC_HB1603_DEP1, DEP3, and DEP4 were post-processed and analyzed for unidentified pygmy/dwarf sperm whales. The 192 kHz recording from deployment NEFSC_HB1603_DEP3 was also analyzed in real-time onboard the ship for presence of Blainville's beaked whales, bottlenose dolphins, unidentified common dolphins, goose-beaked beaked whales, Gervais' beaked whales, unidentified Risso's/bottlenose dolphins, unidentified True's/Gervais', unidentified pilot whales, pantropical spotted dolphins, Sowerby's beaked whales, sperm whales, striped dolphins, True's beaked whales, and unidentified dolphins.

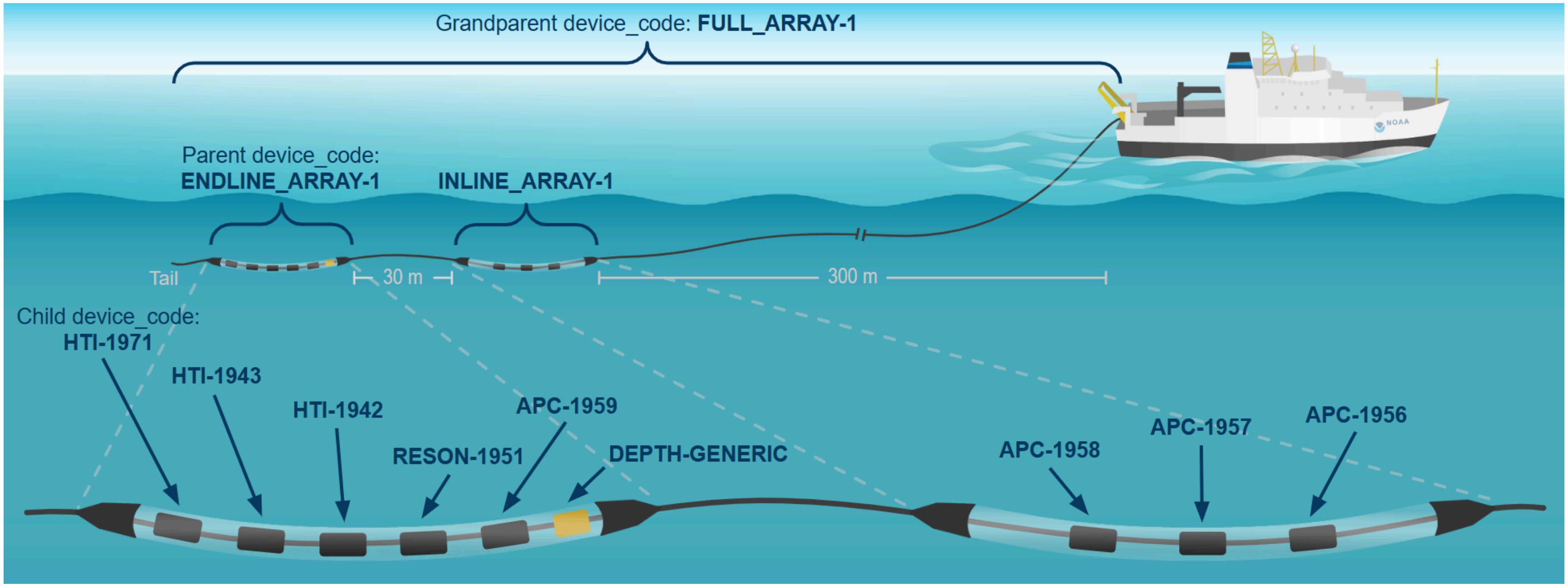

Diagram of a towed array of hydrophones being towed from the back of a research vessel. The labeled, nested devices correspond to the devices listed in the Example 3 templates.

Example #4: Glider deployment

Location of example templates within zipped folder: examples/EXAMPLE_4_Gliders

This example shows an autonomous Slocum glider that was deployed near Cox Ledge off the coast of Massachusetts and Rhode Island from October 2022 to December 2023. The glider was equipped with a second generation Digital Acoustic Monitoring instrument (DMON2) and a Vemco receiver. The DMON2 was recording continuously throughout the deployment with a sampling rate of 2 kHz. Subsampled periods of simplified spectrograms (without the acoustic recordings) were transmitted to shore via satellite every 2 hours, and then analyzed for presence of sei, fin, humpback, and North Atlantic right whales in near-real time . The full audio recordings were collected when the glider and hard drive were retrieved.